

Heat transport in chaotic magnetic fields

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1 Create a map

We want to create a map of the magnetic field lines using PyOculus. The magnetic configuration considered is described using slab geometry through

$$\psi = \psi_0(x) + \psi_1(x, y, z), \quad (1)$$

where ψ_0 describes the equilibrium configuration and ψ_1 is a perturbation of amplitude ϵ with poloidal and toroidal mode numbers m and n :

$$\psi_0(x) = \frac{\ell'}{2}(x - x_0)^2, \quad (2)$$

$$\psi_1(x, y, z) = \epsilon(x) \cos \alpha \equiv \epsilon(x) \cos(my - nz). \quad (3)$$

For a system with $x \in [0, 1]$ containing an island at $x_s = 0.5$, as a simple model for a perturbation localized around the rational surface, we choose

$$\epsilon(x) = \delta \cdot x(1 - x). \quad (4)$$

In PyOculus, all the calculations are performed in cylindrical geometry. For now, we'll consider a simple mapping between the slab and cylindrical geometries, where x is mapped to the radial coordinate r , and y and z are mapped to the poloidal angle θ and toroidal angle ϕ , respectively.

As for the magnetic configuration itself, there is no specific function in PyOculus that reproduces exactly the case we wanted to study.